Release Notes xxxx-xx-xx : : not ready at all
The AD literature seems to go from miracle to failed miracle almost daily with little attempt to integrate new results with old observations. There are now a lot of data poitns to put together with cause and effect like any TV detective would do or as described in fables like the blind men and the elephant even though apparently Natalie Merchant set that fable to music (the subtitle refers to the band 10000 Maniacs also suggestive of a metnal disturbance). This work proposes that all these stalled approaches can be causally explained by well known issues such as age related digestive defects. Suggestions are further supported by in-house work on optimization of dog diets demonstrating apparent safety and utility of out of favor vitamins that could be impacted by changes in stomach chemistry. While most of this work builds on prior results, it does end up advocating dietary changes that may not be entirely obvious from current thinking. Another problem may occur from exploring the obvious suspect tyrosine with its central role in both brain chemistry and skin pigmentation. Often, glib comforting assertions about socioeconomics are advocated in the literature and its not clear if this topic has been meaningfully explored. This immediately jumps out as a suspect in race based AD differences especially given other observations considered herein. To be sure, constituitive skin pigmenters may have adaptive mechanisms which support tyrosine flux with no net "bad" effects elsewhere and there may turn out not to be any anwyay. In any case, its an important issue and like any other law of nature will not yield to human desire and the only way for people to benefit is follow the data. If you want to fool society, appease angry people, instead of solving problems and eliminating run on sentences for everyone turn back now to your safe space...
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Alzheimer's Disease : 10k maniacs agree its an elephant

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(Dated: March 30, 2026)

Effective treatments for Alzheimer's Disease(AD) have been difficult to create despite its well known morbidity. Much effort has been directed at amyloid beta removal with minimal clinical benefits demonstrated. Several distinct hypotheses have emerged that focus on things such as infection, vitamins, other various small molecules, meal or dietary selections, toxin exposure, or GI tract disturbances. Each group of approaches can find a rationale and typically some encouraging early results but no easily demonstrated robust patient benefits. Taking these results and other observations together, similar to the fable of the blind men grasping different part of an elephant, a plausible unifying theme is nutrient deficiency due to age related digestive defects. Isolated profound nutrient deficiencies will not likely exist so obvious attribution of disease to single molecular entity is not a good starting point. A robust solution will need to restore nutrient delivery and mitigate any age related endogenous toxin production. Adaptive and malapative responses may make such an initial cause difficult to discern. The literature on the gut-brain axis and dysbiosis addresses some of these issues but does not attempt to fix specific causes for GI distrubrances. This work hopefully finds a rational basis for meal design and motivates "meal engineering" as an intervention strategy that may be able to correct many of these problems. A "meal" may include traditional foods and spices but implied to also contain a variety of other components such as vitamins or other small molecules to improve performance in the impaired GI tract, If this analysis accurately captures cause and effect in the real disease, correction or prevention may be fairly easy with well known small nutrient molecules or addition of acid or chloride to normal meals. Individual responses to nutrient details will appear to make each case different but still possibly treatable with similar simple remedies. Indeed, there has been some limited success with dietary interventions but these fail to be designed around likely age related causes although they may empiricaly mitigate some of the disease. It may not be possible get all required nutrients from food once damage to digestive system has occured and supplements or other entities may be required. Cations such as metals, protein bound nutrients, amino acids, and lipophilic nutrients would likely be the first suspects. In particular, its likely that vitamin K and copper, along with more accepted components such as amino acids and SMVT substrates all need to be included to replace the younger absorption profile. However, brute force supplementation without regard to formulation may not work due to chemical and metabolic interactions. Two important amino acids with solubility issues in particular, tyrosine and tryptophan, and critical but reactive metals such as copper may require special attention. There are also possible political concerns if there is a hesitancy to link tyrosine metabolism to dementia. However, as always, social influences will not change natural phenomena.

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@techreport{marchywka-MJM-2026-001,
filename ={elephant} ,
run-date ={March 30, 2026} ,
title ={Alzheimer's Disease : 10k maniacs agree its an elephant} ,
author ={Mike J Marchywka } ,
type ={techreport} ,
name ={marchywka-MJM-2026-001} ,
number ={MJM-2026-001} ,
version ={0.00} ,
sourcecode ={https://github.com/mmarchywka/elephant} ,
institution ={not institutionalized, independent } ,
address ={ 157 Zachary Dr Talking Rock GA 30175 USA} ,
date ={March 30, 2026} ,
startdate ={2026-01-17} ,
day ={30} ,
month ={3} ,
year ={2026} ,
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pages ={ 36}
}
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I. INTRODUCTION

Alzheimer's disease [49] remains an unresolved problem with important impact to society [41] despite all the money and effort directed at it for the last several decades [57] . Anti-amyloid products have been the major focus but continue to show questionable net clinical benefiti [216] [194] and do not always get FDA approval even with creative trial design [100] [79] . This is in spite of the fact that as early as 2002 some had noted too many inconsistent observations about amyloid or tau for them to be serious targets [185] . Several other approaches and theories have been developed with various levels of support although it would be difficult to say that a specific next-best has emerged. The thesis of this work is that all these distinct approaches can be resolved by looking upstream for initial causes for these diverse observations about the disease. Before detailing these "pieces of the proverbial elephant", some indication of the problem may be appreciated with recent effort to emply AI against AD. Given the different aspects of AD, often described as "multifactorial" or "heterogenous" emerging AI has been proposed to lead a response [113]. [84]. This may yield results, its progressing too rapidly to rule out, but often it is limited by traning not on raw data but human commentary and a statistical rather than "model of reality" based view. Others are using AI for precision medicine thinking that patient heterogeneity is a problem [260]. Some are searching for connections between apparently unrelated observations [96] to find tractable intervention strategies similar to the present work. Headlines sometimes appear about new scientific or anecdotal evidence pointing to a new treatment but rarely do these translate into useful therapies. Part of the thesis of this work is that many of these results are useful but tend to be interpretted in isolation and without regard for cause and effect in real disease despite often intricate and detailed biochemical anlyses related to each piece. While some pathway details have been described, the search backward to an "initial" cause is not common. Often the actual cause is treated as irrelevant. "Animal models" may mimic some features of real disease but are predicated on a cause unlikely to be relevant to the real disease [201] and their limitation are becoming more appreciated . By coincidence, sometimes however results may be transferable but don't seem to add to understanding. These will be discussed in more detail in the context of each piece.

The following subsections list some AD related theories or data that motivate the fuller thesis of this work identifying a more of less tractable point of failure which leads to diverse symptoms but yet may be easy to treat. In essence, this work suggests that the best intervention is in the GI tract, similar to dysbiosis theories and malnutrition and diet theories, but with the goal of addressing specific causes of these consequent conditions.

A quick note on taxonomy and focus may be helpful. Dementia in general and AD in paricular are not well understood but can be defined by a few concepts. AD is generally defined by amyloid and tau markers [123] but as mentioned later there is reason to believe these have various reasons to occur. Typically AD is divided into a sporadic and familial or late-onset (LOAD) and early-onset (EOAD) type with work continuuing to explore differences [205] [30] [261] . This work is generally oriented towards late onset sporadic AD although its likely to address many age related issues too as nutrient effects are not entirely brain specific. Indeed sarcopenia will be noted as a correlate or age and disease. The implicit feature of AD assumed through out is that it is a disease of older age and theories need to explain that age dependence. The remaining introduction sections discuss classes of observations or data, "pieces of the elephant" , that the present work claims to organize into diverse manifestations of age related GI impairment as detailed later.

I.1. Some stumbling blocks

Thinking outloud
on the one hand, this doesn't fully capture my disguest but its also too distracting as written ROFL

The hypothesis presented here largely builds on prior work rather than contradicting many of the details but there are a few exceptions to that. When proposing a new theory that is inconsistent with current thinking, and prior work is stalled, it may be that theire are fundamental fallacies in the existing accepted analysis or data. To make the new theory credible it may be important to explicitly identify hidden fallacies or errors in the status quo. Some common ones were enumerated in prior work [165] I listed several common reasons for dead end thinking as the literature appeared to suffer from many. Here too many of those apply and the fixation on amyliud beta, treating numbers to be more normal, ignores upstream cause and effect. AD however is also apparently restricted by social factors as intelligence and behavioral issues may elicit more rage than reason from sponsors. Comforting assertions and assumptions, sometimes to gain more popularity, may be held well beyond what is justified by any data. In other cases, cause and effect may be related in implausible ways which corrupts clear thinking. For example, a recent work looking at sex and race as factors in AD suggests that " educational level" a modifiable risk factor [189]. That may be true to some extent but realistically some may have preexisting brain disfunction making it esssentially impossible to achieve meaningful higher education. And we are already seeing unfortunate consequences of thinking a degress itself, without real accomplishment means anything with the distroted job market [71] [81]. Or, maybe all we need to do is give AD patients a PhD in physics.

I.2. Accumulated Damage Theories

One general class of competing ideas involves "accumulated damage" theories where some cellular components develop non-specific irreprable damage leading to the decline of the brain and body. These can eqaute chronological age with "wearing out" or invoke something like senescence [18] and just accept "it happens." These may concentrate on vasculature and boarder on hopelessness [12] or ROS damaged proteins [157]. Accumulated genetic damage [276] could include a related group of theories. It may include epigentic changes [168] , somatic mutations and mosaicism [148] , and telomere shortening [136] . This work addresses the protein quality issues directly and epigenetic changes tend to be programmed responses to things like nutritional status so they are expected to improve too. One work concerned with epigenetic factors recommended supplements of things like B vitamins and s-adenosyl-L-methionine [215]. Telomere length is not directly considered although it correlates with age some associations with AD are expected and a causal role is being explored [55] . Some correlations between length and cognitive decline appear backwards [117]. A large number of nutrients have various correlations with telomere length such as the usual B-vitamins [196] as well as diets [230] such as the Mediterranean [85]. There also appear to be associations with vitamin K intake [59].

Genetic mosaicism is a normal part of the brain as mutations accumulate from conception [23]. Many of the pathological mutations linked to AD are thought to be encouraged by well known AD risk factors [118] suggesting they can be reduced by addressing these known factors. Stabilization or removal of deleterious somatic mutations may be less obvious. Certainly generic defects in DNA repair may not be remediated solely by diet but defect production and culling of mutants may be possible.

It is perhaps important to single out somatic mutations that may lead to splicing errors are splie variants have been linked to aging and AD [156] [227] [144]. This will come up in more detail later regarding splace variants of nutrient transporters. Of particular note, a splicing mutation, " appears to disrupt the inclusion of an exon in MS4A3 by perturbing a YBX3 binding motif, a hypothesis supported by our ASO targeting of this site as well as YBX3 CLIP binding and shRNA KD. Although our result is only significant in blood, other genes in the MS4A gene cluster have been previously implicated in Alzheimers through blood [64]. This is most likely due to monocytes in blood acting as a proxy for microglia." [256]. YBX3 posttranscriptionally controls a transporter, SLC1A5, that apparently is important for intacellular amino acid regulation in skeletal muscle [16]. Other works also mention a role in other transporters SLC7A5 and SLC3A2 [51] extending its significance in amino acid control. This fits nicely with the thesis of this work addressing nutrient limitations in old age.

I.3. Genetic Causes

Ultimately all disease is genetic since perfect genes would anticipate and deal with everything that can occur but realistically these consider less common mutations that differ from the larger population. The genetic damage theories have some appeal because several gene variants have been associated with AD and appear in pathways that are plausibly causal although this may create confirmation bias in "relevant" gene selection. One recent work notes hundreds of genes have been implicated and points out that amyloid beta could be a response to stimuli such as, " including infectious agents (herpes simplex, chlamydia pneumonia, and Borrelia burgdorferi) as well as Vitamin A deficiency, hypercholesterolaemia, hyperhomocysteinaemia or folate deficiency, oestrogen depletion, cerebral nerve growth factor (NGF) deprivation, diabetes, cerebral hypoperfusion (leading to hypoxia and hypoglycaemia) " [38]. Then the above concentrates on immune related genes. As "genes" are nominally constant through out life, their roles in age related disease may be indirect such as through reduced DNA repair abilities allowing for damage to accumuale faster with age. For the present theory, its worth noting that no popular GI related gene variants have emerged.

Genetics have been widely explored with several common mutations linked to either familial AD(PSEN1,PSEN2, APP) or sporadic late onset (APOE) disease but with many others possibly involved [13] . Among these are things like SLC10A2 and a variety of lipid, calcium, or immune related genes. APOE is associated with many vitamins including vitamin K. APOE was identified as important for vitamin K transport to bones as early as 1996 with apoe4 noted for association with lower serum K and increased fracture risk [134] and it was noted response to supplementation occurs over months. However, a later work found "superior vitamin K status is associated with the apoE4 genotype in healthy older individuals from China and the UK" [266] .

Thinking outloud
find the work on secretase and hair color for example

A some human mutations identified in specific populations, along with humanized genes, have been used to create animal models of AD. The 5xFAD mouse overexpresses mutations in APP and MSEN1 and mimics some features of real disease although not NFT's [191]. EFAD mice include a human APOE gene [242] but still interpretation of any intervention has to be tempered with an understanding of the assumptions inherent in these models. Interestingly, the 5xFAD mice respond differently to diet than wild type. One study found some apparent decoupling of results,

> Thus, 5XFAD mice exhibited greater susceptibility to highfat diet than wildtype mice regarding cerebrovascular injury and cognitive impairment. On the other hand, 5XFAD mice fed highfat diet exhibited much less increase in body weight, white adipose tissue weight, and adipocyte size than their wildtype counterparts. Highfat diet significantly impaired glucose tolerance in wildtype mice but not in 5XFAD mice. Thus, 5XFAD mice had much less susceptibility to highfatdietinduced metabolic disorders than wildtype mice. [152].

These models were largely designed with the assumption that accumulation of amyloid beta was the central feature to replicate but that may turn out to not be correct. Since the cause and effect in real disease would only be captured by luck there could be a lot of limitations from any of these models. Interestingly, transgenic mice may show intestinal symptoms correlating with brain pathology [161].

AD risk varies significantly by sex and race [151] making the related genes suspects. However, other factors such as sociocultural [80] "unique social and environmental pressures" [180] are often blamed. Correlates of sex such as estrogen production may be a more direct factor [209].

Some polymorphisms of the digestive tract are thought to significantly alter drug disposition and they can be race related [88].

More generally, there could be important race or sex based differences in many catagories that are due to easy to treat malnutrition and a focus on social factors distracts from a real solution. Vitamin A related polymorphisms were found to correlate with ethnic group of vitamin A status in a sample of pregnant women [240]. One study of DIO2 polymorphisms and AD among various groups demonstrated a possible contributor to AD in blacks along with a lot of transcription data [174] that may be a good starting point for further hypotheses.

In fact, more directly relevant genes have been found in the black US population. A dopamine receptor D2 variant (DRD2 Taq1AA1) more common in blacks than whites likely associates with AD risk [27]. Presumably variants would differ in some binding kinetics including to dopamin analogs that could become important as parent neurotransmitter becomes more scarce with an aging difective tract.

I.4. Clues from Transcriptional Associations

Besdies mutation data, transcriptome analysis has found genes which are over or under transcribed in Alzheimer's Disease. As with most data classes, this has not created an unambiguous path to an intervention and cause and effect, good or bad, remains unresolved for most of these genes and their products. A gene may be over or undertranscribed depending on how it is regulated. Functional feedback could signal for more of a dysfunctional product to be created or one that can not function for some system related reason. So, typical interpretations may be missing important bpoints that may fit well into the model proposed herein.

A 2025 article found ADAMTS2 to be one of the genes robustly increased in black and European Alzheimer's brains [247]. Essentially this translates into pervasive "weak structural proteins" as described below. Mice with non-functional ADAMTS-2 develop fragile skin [146]. It is over expressed in mice and humans with hypertrophic hearts and this condition is worse if ADAMTS-2 is lacking [258] The information so far suggests it is upregulated in cases where is may be beneficial but is unable to accomplish whatever the feedback is regulating. In the case of structural proteins, something may be signalling " more or better " but another factor is lacking (ie malnutrition).

A 2021 work described a liver-brain axis and implicated junk food consumption, perhaps decomposed into both nutrient and toxin exposure, as a driver of AD with related changes in expression of genes such as, " TUBB5-HYOU1-SDF2L1-DECR1- CDH1-EGFR-SKP2-SOD1-IRAK1-FOXO1" and corresdpondingly, " [gene signature in the] decline of concurrent liver and brain health. Dysregulated levels of these genes are linked to molecular processes like cellular senescence, hypoxia, glutathione synthesis, amino acid modification, increased nitrogen content, synthesis of BCAAs, cholesterol biosynthesis, steroid hormone signalling and VEGF pathway." [110].

I.5. Infection Hypothesis including Cumulative Damage

Another approach gaining interest is the infectious hypothesis of AD [97] [32] [175] [35] . The basic theory is that one or more pathogens cause AD although many are very common such as P gingivalis [93] , varicella zoster [34] , COVID-19 [17] , the Lyme pathogen [97], and Chlamydia pneumoniae [40]. With one abstract listing several diverse contributors [199],

> viruses (e.g., Herpes simplex type 1, 2, 6A/B; human cytomegalovirus, Epstein-Barr virus, hepatitis C virus, influenza virus, and severe acute respiratory syndrome coronavirus 2, SARS-CoV-2), bacteria (e.g., Treponema pallidum, Borrelia burgdorferi, Chlamydia pneumoniae, Porphyromonas gingivalis, Prevotella intermedia, Tannerella forsythia, Fusobacterium nucleatum, Aggregatibacter actinomycetemcmitans, Eikenella corrodens, Treponema denticola, and Helicobacter pylori), as well as eukaryotic unicellular parasites (e.g., Toxoplasma gondii) may factor into cognitive decline within the context of AD.

Part of the motivation is the lack of a clear consistent pathology related to amyloid beta but an apparent role in the brain's immune response [107] (which also lists many organisms and advocates infection control) [87]. It has also been observed that H pylori infection associates with prevelance of AD [195], an observation considered by the present work to indicate a related vulnerability is causal to both. Neurosyphilis can be difficult to distinguish from AD [178] creating precedent for at least and AD mimic.

In further support of this , a wide range of vaccines has been found to reduce risk of AD or other dementia [159]. Another study found benefits of vaccination and also anti viral treatment for herpes zoster [167]. There is also some indication AD may be transmissible similar to prion diseases [223] [31] .

Thinking outloud
microcyline and stroke could be stopping hypoxic virulence from unknown pathogen

The observed age dependence of AD compared to life long exposure to pathogens has to be reconciled with something like a cumulative damage scenario or an age-dependent host response. Many pathogens cause known GI damage that is often ignored. Covid-19 GI damage and interaction with ACE2 receptors has been studied [169] and secondary bacterial infection with gut organisms is common [21]. Varicella zoster similarly can cause GI issues [184].

Some organisms trigger an interferon response leading to the loss of nutrients like tryptophan [272]. Although interestingly an analysis of transcription factor RNA in amino acid deficient bovine mammary cells indicated that the deficiency could trigger interferon signalling as well as re-regulation of pathways related to proliferation and inreased expression of hypoxia inducible factor 1a [142]. This illustrates problems in finding original causes from a complex set of associations.

Nutritional immunity can be invoked [186] towards various nutrients with particular emphasis on some minerals like copper [47] that may redistribute in ways that become pathological if deficient. While amyloid may have prion like properties, this kind of transmission does not seem likely in the vast majority of cases. Hypermetabolism has been described in Covid 19 although perhaps underestimated [33] .

I.6. Gut-Brain Axis Hypotheses

Another general group of hypotheses involve the gut-brain axis [111] [114] [145] [269] [198] in which something wrong in the GI tract drives AD. The theory presented here has a similar origin but will deviate from all of these theories by focusing on specific reasons for the observed perturbations.

A gut-brain-axis has been discussed which includes a diverse range of mechanisms for exchanging information between GI tract and brain [245]. A change in microbial metabolites, many able to reach the brain, is one important component [58] which is shared by the present hypothesis.

Typically this points to something such as the microbiome that gets "out of balance" and needs to be adjusted with innoculation or some feedstock mix or fecal transplant. Some dysbiosis works recognize an association between AD GI tract disorders and then tend to suggest things like," antibiotic therapy, probiotics and prebiotics, and faecal transplantation are being considered as possible therapeutic options" [43].

Often nutrition is considered and a detailed analysis of cause and effect between gut and brain is performed. This may lead to similar nutrients as those consisdered here but doesn't usually get to the initial cause hypothesized here nor design a solution around that cause. A variety of causes for dysbiosis have been considered but most tend to be artificial. Aged transgenic mice administered a 3 antibiotic combination for 14 days displayed dysbiosis and imparied cognitive skills along with various other distrubances [248].

Various things like phytosterols have been proposed to fix dysbiosis [86] with some preclinical encouraging results although ignoring actual causes.

I.7. Metabolic theories

Many of these relate to nutrient issues but I guess can be treated separately. These form a wide range from very general observations about hypometabolism and dysregulation including insulin resistance [239] to various named catagories. An "Inverse Warburg Effect" proposes that AD results from upregulation of mitochondrial oxidative phosphorylation, the opposite of what is observed in cancer [64]. Although the stated cause is " natural result of the aging process and derives from the cumulative effects of random perturbations of the cellular components involved in energy metabolism " the above work suggests increasing lactate production and moderating ROS generation. In essence though this is a cumulative damage theory. Others are based on insulin resistance [74] and all the factors that effect this condition in the periphery and relationship to metabolic syndrome [257] . Indeed, the basis of hope that GLP-1's will be useful in AD is the issue with glucose utilization [147] [101]. GLP-1's will be discussed in more detail as the present theory is described as the problems illustrate nutrient issues in aging. Another cause of decreased metabolism in the brain is "age-related decline in the ability of glucose to cross the blood-brain barrier "[26]. Vascular and metabolic issues may be combined in other works [108]. Other theories suggest evoluationary programming of an age related metbolism [213] which presumably is encoded in common genetics. The metabolic changes are explained in the present theory as age related nutrient limitations on barrier and transporter function.

I.8. Nutrient Hypotheses

This work will consider many combinations of nutrients that have previously been identified as possibly useful in AD. However, these prior works generally don't try to determine a cause or a reason to single out one or a few nutrients. Some of the catagories are discussed here to illustrate the concepts of this theory group while others are discussed where the present hypothesis is presented. A more coherent presentation will bring these together as specific combinations are considered probably in a later work.

Works cataloging the utility of vitamins or derivatives for AD exist [252] [226] but often treat these things as "one at a time" approaches. Although at least one review from 2022 recognized the likely need for multiple components [6]. Combintations often combine 'usual suspects." Many combinations appear to exist for B vitamins, anti-oxidants, and other trendy things. Just this year a work suggests combining the usual suspects A,Bx,D,E,K [182] apparently neglecting protein and minerals and without real rationale. Another work combined A,D,E,K and multiple B vitamins as well as probiotics with motivation being oxidative stress, inflammation, and dysbiosis [238]. However, its not clear what is causing all those things or why they matter. In particular, oxidative stress may be a signal for among other things copper deficient mitochondria. Often mixes seem to be predicated on old ideas such as a combination of cartenoids, omega-3 fatty acids, and vitamin E which also may demonstrate a passing benefit [190]. The point of this work is to determine, or at least guess based on current knowledge and observations, a better cause to focus the response. Some mixes such as a mitrochondria nutrient mix [155] have been considered. Too many to mention employ some combination including obscure or exotic herbs or ingredients [124] often for glib reasons. Some success has been observed in probiotic trials [197] which of course likely effect GI chemsitry. Dietary patterns, similar to issues with "meals" discussed here, have also gotten some attention for empirical correlation with dementia such as a MIND pattern [53], Medietteranean and DASH [259], Japanese [274], and there are others [228] with various outcomes considered. A high sugar dietary pattern was analyzed with HR as a function of sugar intake showing an optimum but increasing beyond that [275]. This will be useful later if considered in the context of auto-brewery syndrome and methanol exposure. The partially beneficial ones of course suggest ingredients for a more complete solution but many details and false leads may need to be sorted out.

I.9. Various Toxin Hypotheses

A variety of theories exist about cumulative exposure to environmental toxins such as "harmful metals, pesticides, agrochemicals, and air pollution " [65] or " heavy metals, flame retardants, and pesticide residue" [90]. Agent orange may produce some similar symptoms as AD [61]. This work will look mostly at toxics thought to be produced endogenously in the impaired stomach. The most prominent of these is methanol and its potential metabolites such as formaldehyde. Methanol has been well explored for cognitive impairment [143] as has formaledhyde [267]. These will be discussed in more detail as the current hypothesis is elaborated. However, a huge array of induced secondary metabolites are possible from any GI tract microbiome so its difficult to rule out anything. Some potentially important ones have been discussed [277]. Conversely, some marine dinoflagellates toxins have been considered for reduction in amyloid burden [29]. And its also worth mentioning that acetylcholinesterase inhbitors while usually considered toxic may be therapeutic [232].

II. THE PRESENT HYPOTHESIS

Thinking outloud
this needs a tractable statement in once place and no point in cluttering with citations here. This is the "tell them [again] what you are going to tell them part ."

The point of intervention presented in the present work of course relates to all of the above "pieces of the proverbial elephant." It recognizes that nutrient deficiencies likely exist and that metabolic and gut distrubances such as insulin resistance and dysbiosis exist as the result of reduced nutrients and or increased endogenous toxin product due to aquired GI defects. It is these earlier defects that are the target of proposed interventions rather than the downstream effects thought to be diverse symptoms of this more tractable cause. Many nutrient-toxin problems though may cause maladpative responses over time and simply changing the balance back may not be sufficient if defects in transporters for example have been introduced. Unfortunately some adpative or compensatory responses may be sereptiously auto-catalytic. Nutritional immunity, or the hiding or destruction of nutrients, may simply create more vulnerability to infection and more of this reaction if the lost nutrients can not be restored. So, some downstream interventions may be possible but simply forcing numbers like blood sugar to normal values may not be the best response although sugar itself can be destructive it depends on any collateral damage from reducing it. While much empirical data needs to be collected, an early intervention may be as simple as engineering meals with modified components similar to those already suspected of reducing AD risks. Adding chloride along with other spices may be enough to aid digestion and decrease toxin production. Later, supplements may be more effective as digestion is too impaired for normal food to work. More obscure or less obvious small molecules may be needed later for various reasons. So, the interventions should look at lot like those in nutrient or dysbiosis theories but with different specifics due to focus on a specific cause not restoration of normal numbers or microbiomes.

Thinking outloud
this will never get out the door unless specifics are deffered to a later work..

A 2025 work did elaborate on the concept of pre-digested food for special populations such as the elderly with a detailed review of some issues [262] and it discusses some approaches such as fermentation that may be explored herein. However, it does not appear to address toxin production by the distressed GI tract.

As no particularly good suspect has emerged, it may be worth looking at all the data and trying to guess at age related vulnerabilities that cause infections to lead to dementia, digestive disturbances, and classic symptoms such as amyloid build ups. Nutritional defects make some sense as frailty, more than age per se, may be a correlate of age related diseases.

As in the prior theory description in the introduction, the following will elaborate on specific small molecules, or classes of data or observations or theories that more directly relate to supporting the present hypothesis.

II.1. brain microbiome

In a prior work [164] I interpreted a brain microbiome result to describe some host factors that would explain differences in the most and least abundant organisms in AD and healthy brains post mortem. Microbiome studies on low-biomass samples have been criticized previously for susceptibility to various errors [130]. While the existence of a brain microbiome is controversial due to the expected low absolute abudance and opportunity for contamination, the analysis is somewhat robust in that contamination from other parts of the patient should still reflect differential body nutrient levels. One argument against contamination was similarity to an unrelated work on uterine microbiome and possible symbiotic role of these organisms but still a lot of coincidences can occur.

Thinking outloud
I'd mentioned this before but it may be worth trying to stably infect a uterus with labelled genetics and look for the inseted gene in offspring to help distinguish from external sources. Something like GFP would be cool but there is no assurance it would be expressed as there is a concern about CNS specific adaptations but even trying to find these organisms in an offspring uterus may be informative if raised apart from parents etc.

This work generally pointed to deficiencies in lipophilic amino acids but also suggested additional methanol in the AD patients. One likely source for this is the GI tract due to changes in microbiome or initially changes in chemistry. Other toxins could associate with methanol production too and a good strategy may be to restore GI chemistry back to youthful state.

II.2. methanol and auto brewery syndrome

Methanol can be generated in the body from ethanol [66] while both ethanol and methanol circulate in a normal person at the .5-1.5mg/L level and may be produced from carbohydrate consumption [115]. Endogenous ethanol production in the stomach has been observed [236] and considered. One post mortem found high ethanol in the stomach attributed to Candida glabrata [138]. A recent review of auto-brewery syndome suggested specific organisms may not be the issue as much as "...the use of medications, notably proton pump inhibitors and antibiotics. When combined with a Western diet rich in fat and simple sugars, these factors significantly contribute to altering the gut environment and subsequently, ethanol production." [177] although a list of ethanol producers has been created [173]. In one patient an oral glucose tolerance test achieved an ethanol concentration of 315mg/dL in the blood [225].

II.3. Neurotransmitter precursors etc

Thinking outloud
move some of this to the specifics sections later

Amino acids are thought to play a role in AD but several results suggest them to be unimportant largely probably due to glib or superficial interpretations. A 2026 review of neurotransmitter systems in AD noted that glutamerigic and GABAergic systems tended towards a more excitable state while histamine levels were reduced while cholinergic, serotonerigc, dopaminergic, and noradrenogeric systems showed more degeneration [210]. This state could be created with limitations in essential amino acids and choline combined with excitatory compensation in the other systems.

II.3.1. Choline

Choline has been the topic of several recent works [60] [254] adding to a long history for it [253] [44] and related pathways [82] including cholinesterase inhibitors [22]. The 3xTg-AD mouse apparently has low levels of circulating choline [116] and that may be instructive to determine if the model genetics relate to this. While often related to acetylcholine, its important to remember its possible role in nutrient absorption although in a later section on digestion it was determined that under non-physiological conditions but in a digestion related task choline chloride acted purely throught the chloride and was equivalent to salt [233].

II.3.2. tryptophan

Tryptophan is a good suspect for many reasons but various supplementation trials have yet to show an interesting benfit in AD. In this work, the failures will be attributed to formulation, a need for other componets of the limiting pathways, and failure to address any sources of toxins like methanol from the pathological GI chemistry which may exist.

Among treated metabolic syndrome patients, the Kynurenine to tryptophan ratio, as well as inverse C20:3 and lysine amounts, correlates with the risk of mild cognitive impairment [112]. This work also mentions patient heterogeneity which may be an important factor in testing nutritional interventions.

Tryptophan has as a contributor to many diseases has been reviewed in several publications [200] [119].

A mixture of 9 essential amino acids includinc tryptophan given for 28 days failed to show any meaninful benefits [243] but this may be toolittle, too short a time, the formulation was not clear, and given together they can compete for entry into CNS. Other factors may be needed to get uptake and transport. A tryptophan dipeptide, a product of fermentation which may have improved absorption, did have some effects in mice [14]. At 25mg/kg BW for 12 weeks, elderly with mood disorders showed some improvements [48]/ although formulation was not clear. A variety of tryptophan derivatives have been explored such as 5-HTP [4] and indole 3-propionic acid [135] have also been explored.

Once nutrient issues occur, they can cause other problems such as transport reduction making simple supplementation less likely to work easily. One small study in rats found an age dependence of transport across the BBB both saturable and passive diffusion rates decreased in older animals [244]. This would suggest that at a constant blood level less enters the brain and supplementation with trytpohan may be futile. However, the transport restriction may be simply due to loss of other nutrients with age. Of particular interest was the 1991 observation that oleic acid can open the blood brain barrier [241] which has been observed to change with age and AD [212]. The quality of the BBB could then be responsible for many nutrient deficiencies in the brain but as with other structural components may be limited by nutrient uptake from the GI tract. Note too these issues are not mutually exclusive. Tryptohan occurs as a conserved amino acid in transporters and at membrane interfaces. Once tryptophan is depleted transport and barrier may suffer.

II.3.3. histidine

Another boarderline amino acid may be histidine which is precursor to histidine. One study in mice claimed it improves response to chronic hypoperfusion [234] although data suggest small differences in quantifies explored. A study a few decades ago did find reduced regional histidine in post mortem AD brains [172]. At least one clinical trial in AD was considered, NCT06169826 https://clinicaltrials.gov/study/NCT06169826 but apparently withdrawn.

It may provide protection against ischemia [150] and reduced oxygenation may be a factor in aging.

II.4. benzoate

Sodium benzoate has been explored as a therapy for dementia apparently due to D-amino acid oxidase inhibition [154]. D-amino acids constitute a set of signalling molecules only recently appreciated as possible contributors to AD and other conditions [62] so such an MOA would be plausible. However, this work also noted that sodium benzoate can act against various pathogens as a substitutde for things like chlorine [46] but it may have a more complicated role in microbiome and may be produced endogenously [98].

It may have a shown a benefit for women in early phase AD and also increased some hormone levels [153]. A more recent review suggested it may have an effect in various mental conditions that is likely stronger in women than men [149]. Another review cautiously suggested a benefit in early stage AD but noted the small sample size and reliance on small number of trials [162]. It may only function in cases where dementia would otherwise be dominated by methanol or some other fermentation related products. Studies have been conducted on sodium benzoate and the GI microbiome largely due to concerns about toxicity as a food additive and results have been mixed although apparently well established for benefits in farm animals [141] [133].

II.5. arginine

Oral arginine as recently described as suppressing amyloid pathology in animal models of AD [83].

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II.6. glp-1

GLP-1's have finally been getting popular press attention due to related nutrient deficiencies such as scurvy [231]. In at least one case report, with no suggested GLP-1 usage, severe GI symptoms may occur from reduced retroperitoneal fat pad buffering the aorta and superior mesentaeric artery combined with other problems like scurvy [9].

And indeed scury, another state creating "weak structural proteins"

> **Thinking outloud**
> my new catch phrase, probably including the collagen scaffold which supports appatite deposition

has been noted in the literature in particular with colonic pseudoobstruction [68]. Although there has also been concern raised about under estimating the harm of malnutrition related to their use [75]. Other nutrients of concern include vitamin D, iron, calcium, protein, B-1 and B-12 [250]

Particular concerns have been raised about the hidden danger of lean muscle mass loss and resulting lack of resilience. [249]. The GLP-1 data may also motivate an interest. GLP-1's apparently are causing non-specific nutrient deficiencies maybe similar to problems common in old age. Recent clinical trials showed either no benefit in AD with the Novo Norodisk product [2]

> **Thinking outloud**
> does the author realize running against "other" antidiabetic drugs is misleading? ROFL

or a positive trial [72] appeared to be cherry picked. A Novo-Nordisk presentation on the results, https://sciencehub.novonordisk.com/content/dam/sciencehub/global/en/congresses-and-scientific-publications/congresses/ctad2025/johannsen1/sliders/Symposium_4_Cummings_Jeffrey_V2_OH.pdf does suggest largely no effect either good or bad although some of the side effects may be unpleasant. Its likely that "early stage AD" implies existence of important nutrient limitations have already been there for years so the differential impact, despite GI symptoms, may be minor.

A pooled analysis of 4 trials and 112 AD and MCI patients showed no significant outcomes differences and numerical harm well below significance [192].

The emergence of clinically important aged appearance (" Ozempic face") [188] [220] and concerns about neuriological adverse events [45] or "brain fog"i anecdotes [246] are well known despite several works suggesting they should be brain beneficial

. There are a lot of lessons here but the immediate thing of relevance is that nutrient limitation that is can be "plausibly denied" appears to generate some symptoms of aging and more to the point frailty. Interestingly, at least one study found a correlation between perceived face age and dementia risk [265].

Evidence pointing towards a cognitive benefit for GLP-1's may be a bit misleading. In one case, they were run against sulfonylureas which have known relevant side effects. In other cases, words suggesting a cognitive benefit in actuality mean that some risk factor has been reduced without clear evidence of real clinical benefit against dementia.

II.7. race and sex

Women and blacks appear to be at higher risk of AD than European males. Factors unrelated to the disease per se may confound results but that is the case with even politically correct association studies. A small 1996 study found that blacks has a higher found higher pH and HCO3 secretion as well as higher prevelance of Helicobacter pylori infection and gastritis [56]. Difference in "normal" protein digestion in healthy males and females appear to be known and have been considered already for a basis of "engineering foods for women" [139]. Note that this empirical difference does not rely on a specific reason although the chemistry may be informative for meal design. A small 1982 study of relatives found that older and female probands had lower acid output although some results depend on how the output is normalized [129]. One study comparing healthy 23-42 year olds with 44-71 found more acid secretion in the older group [91].

While one study did find differences in serum tyrosine levels between older people with and without AD [158] blood levels may vary due to transport issues.

One obvious issue with skin pigment is tryosine metabolism. Both the depltion of tyrosine and generation of toxic analogs could be considered. It is possible that constituitive skin pigmentation (CSP) evolved with adaptive mechanisms that may be instructive to investigate. CSP evolution is usually described as an issue of sun exposure with damage and vitamin D synthesis being driving forces in tone [109].

While CSP may predispose to various maladies, it may also be easily treated if existing adpatations are not already active. Indeed, speculation on beneficial tyrosine analogs can not be dismissed.

And melanin itself is not inert once formed especially in the presence of UV light [235]. A link between dementia and melanin has been considered by many authors. A study of several thousand Taiwanese demonstrated a hazard ratio of over 12 for AD in vitiligo sufferers versus control (HR of 5.3 for any dementia) [42]. Vitiligo is an autoimmune disease making interpretation difficult but it does suggest an age related problem with melanin due to covert pathogen or defective products.

Fringe ideas in the mainstream litearture include ability of melanin to transfer light energy to the cell for ATP creation and loass impairs function [20] . More plausibly, loss of something like tyrosine or copper limits pigment synthesis in some idiosyncratic manner or more to the point consituitive pigmentation requires resources that are no longer plentiful.

In zebrafish, there was an unexpected link between PSEN2 and pigmentation [94] of unknown significance.

Tyrosine anecdotes around menopause may be one indicator. While not considered essential as it can be derived from phenyalanine, it is possible that overall supply of the precursor and regulatory systems reduce tyrosine production pathologically. This is most likely during times of "stress" and shortages. Interestingly in connection with GLP-1 there is a tyrosine dipeptide apparently acting as a signalling molecule Amino acids such as tyrosine may have limited availability in cell cultures and dipeptides have been explored [121] for many years now. Also of interest is the need to supplement tyrosine as a dipeptide to Chinese hamseter ovary cells in bioreactors [255] with work showing possible pickiness of cells for uptake although translation to human GI tract remains to be done. Similar problems have been noted for tyrosine usage in humans from either IV injection or parenteral feeding [160]. Tyrosine-tyrosine peptide is considered a hormone and may be responsible for anorexia in urea cycle disorder patients [181].

Comparison to other racial groups does not appear informative as data are all complicated.

Some context for phenylanine to tyrosine conversion as part of the larger protein turnover dynamics has been explored [89].

Indeed GLP-1 is often considered with PYY for understanding obesity signalling [63]. While its unclear if PYY can replace the GLP-1 drugs, if supplementation is attempted it will benefit a lot from recognizing the issues explored here. Dipeptides or reproduction of "best' GI chemistry may be more important than simply adding tyrosine to diet.

A variety of nutrient deficiencies have been observed to correlate with AD risk such as B,D, and E [50].

II.8. Copper

Copper has a lot of motivation behind it but its rich chemistry may make it one of the more difficult to study. Copper is difficult to characterize but some results may motivate its importance. Overall it appears that AD brains are copper deficient and well formulated supplementation may be beneficial but it can not be excluded yet that other nutrients are limiting its distribution. At least one case report exists of a defect in ceruloplasmin resulting in chorea and dementia with iron deposition that resolved somewhat with fresh frozen plasma [270]. As early as 2016, copper overload was suggested for regenerative medicine and boosting mitochondria performance[217].

Sometimes there is a tendency to think of local levels as a passive result of supply but often complicated signalling generates overall results. As early as 2009, the paradox of excess copper with amyloid beta coincident with tissue copper deficiency was recognized and explained to some degree [105]. In prior work with dogs, the inability of chelation to produce clinical benefit in animals with hepatic copper accumulation pointed to a bottleneck in transport rather than a global excess[165]. Also in the above work it was noted that "copper toxicity" varied widely with the background diet suggesting it has to do with chemical reactions in the meal. Rats raised on a copper deficient diet showed increases in adrenal dopamine-beta-monooxygenase mRna, protein levels, and enzyme activity in decreasing order [204]. This suggests that feedback mechanisms are attempting to increase activity levels with greater transcription rates but unable to metallize the extra translation products. A 2017 mass spectrometry study on post mortem brains found overall decreased copper in AD brains similar to that in Menkes' Disease [264].

One interesting in vitro work was able to image intracellular Cu resolving oxidation states [229]. The work suggests amyloid beta can act similar to a siderophore and import copper, instead of iron, into the cell leading to death unless rescued with a chelator. The authors note the probe may perturb physiology and that Cu is released into the synaptic cleft. Taken together with existing literature, this tends to point to a bottleneck that may be partially neuron specific. Eliminating the excess copper may prevent death in vitro but may not restore normal brain function.

II.9. Vitamin K

For whatever reason, vitamin K is often ignored even though interactions with APOE and appearance in "healthy" foods motivates at least an association with health.

An early proposal for vitamin K deficiency was described in 2000 citing "regulation of sulfotransferase activity and the activity of a growth factor/tyrosine kinase receptor (Gas 6/Axl) " [7].

In the last few years many works have explored vitamin K in old age conditions [73] [120] including association between vitamin K intake and cognition [28]. As bleeding or microbleeding is an issue in AD, its worth noting that 82 percent of "chronic stroke subjects" were vitamin K intake deficient and excessive vitamin E may also be a common problem [92]. The above work also describes more direct effects of vitamin K in AD including interactions with amyloid beta. (there are works that concentrate on emerging problems with vitamin E supplementation too, for example [128])

II.10. NAD+

NAD+ deficiency has also been explored in AD [104] and other age related degenerative conditions [140]. Disturbances of NAD+ can influence trajectory of viruses such as Japanese Encephalitis Virus in mice and human cells [125].

II.11. Taurine

Taurine has been proposed as possible disease modifying supplement [103].

II.12. SMVT Substrates

Biotin and the SMVT substrates [221] have also come up in AD work and dementia more generally [224] [52] . Biotin is critical for many processes and can be obtained from digestion or intestinal microbe production [122]. A 2026 study of dietary biotin and AD risk combined with neuropathology results in biotin deficient mice suggested a biotin decieicny as a contributing factor to AD [273].

II.13. Thiamine

Thinking outloud
doh, don't forget riboflavin stability issues even if not known to be AD relevant

Thiamine is also being considered [77] and that too is through to have acid sensitive absorption [137]. A 2026 work identified a new mechanism by which it could mitigate problems with amyloid beta by suppressing HIF-1a [11]. A 2012 study did show some benefits from supplementation in the elderly but due to poor absorption parental dosing was mentioned [208]. It seems that many studies on isolated nurients have come to similar conclusions but not attempted to think through cause and effect more generally. A 1988 study claimed to use a niacinamide placebo [25] which motivates another problem of an unintentionally active placebo. And in fact niacinamide has been discussed as a treatment too [132].

II.14. Lithium

Lithium has recently been rediscovered [15] although positive review articles date back to at least 2012 [78] or so [170] and its not clear what happened over the intervening decade.

In 1971, it was observed that lithium could impact carbohydrate generally and inositol status in the brain [8]. In another manuscript considering inositol as a dog supplement (unpublished result)

but as of 2005 it could not be determined if lithium or inositol was more directly causal to brain changes that matter in the clinic [95]. This is likely a recurring issue in Alzheimer's and medicine overall. Associations are difficult to map to an actionable cause and effect sequence that can be beneficially modified. It is the main theme of this work that the primary cause is well removed from these associations and the elephant is not apparent from the sum of its parts. It is likely that both inositol and lithium are impacted by some common cause and simply supplementing one or the other or both can make some improvements but is a deadend to a cure.

II.15. Bicarbonate, Chloride, Phosphate

There may be some intersection between detergent chemisty and conditions needed in the GI tract to absorb vitamins and retard toxin impact.

Potassium chloride, unlike other halogen salts, appears to stimulate production of ochratoxin A on wheat [237], /enzoat

III. COMMON AGE-RELATED DIGESTION DEFECTS

To begin to make sense of the above and look towards the design of a better intervention, some knowledge of likely defects and resulting deficiencies needs to be established. Existing literature is a guide but like all published works needs to be considered in light of available data. Most works tend to minimize the importance of typical GI aging on health apparently mostly based on assertions and assumptions. Contradictory observations would need to be considered carefully and its likely intervention design will be somewhat empirical hoping to gain information as work continues. For example Chapter 8 in a 2014 book [126] suggests a large reserve capacity and no real effects absent disease with lowered uptake being a consequence of reduced protein mass. In what follows the "reduced protein mass" may be considered a problem in itself consequent to reduced nutrient uptake for reasons yet to be determined for this work. By 2025 the problem of protein uptake seems to be recognized [99] [207]. Cause and effect may be quite difficult to trace however. Consider the case of aginng colonic muscles cells with reduced performance related to L-type Ca currents [36]. Resting $[Ca^{++}]$ is remarkably low with a large ratio to extracellular levels [127] and therefore sensitive to leaks. In fact as early as 2003 reduced response and intracellular accumulation were considered as a reason for "age-related learning deficits" [171]. Calcium leak may occur for several reasons [163] [166] and mutation related disease suggest problems with high-inflidelity translation that could occur with amino acid imbalances.

By 2025, methods to simulate elderly digestion in vitro were discussed because age-related GI properties were thought to be important for health related outcomes in the growing elderly population [263].

Thinking outloud
probably a bad citation, low quality journal doi info missing lol [1]

In extreme cases such as GI resection, problems related to malnutrition such as increased abscess risk and reduced ability to produce collagen for wound healing and other needs [1] . ADAMTS2 upregulation in the case of AD was part of the motivation to create this work as it suggests a chronic problem in collagen quality.

By 1992, common themes in impairment reports were emerging. One review suggested decreased taste and hypochlorhydria with decreased uptake of B-12 and minerals but increased absorption of lipids [218]. A small 1997 study of independently living people over 65 found about 90 percent had basal pH under 3.5 [106] although there is no indication of how this compares to the young or very old.

A study of 79 healthy elderly people found some indications of lower acidity than young people but did not consider it relevant in most individuals at least for drug absorption [219].

The pancreas and intestines may also be effected. One review mentions decreases in calcium, zinc, and magnesium but not copper in old rats [102]. However, the pancreas and intestines must work with whatever the stomach produces with the meal. Earlier, I considered that lack of an early low pH step may produce more idiosyncratic results as some mineral can irreversibly precipitate [165].

Around 2004, it was thought that reserve capacity was sufficient although some reduced absorption of vitamins such as A,D,K, and B6 but tended to minimize significance for treating aging [69]. One problem however is that even today sufficient intake and uptake may not be clear and the impact of multiple "low levels" could invoke adpative responses.

Today however protein deficiency appears to be reasonably well accepted as an issue with aging and its manifestation as sarcopenia [206]. What mey be less apparent is any changes in quality such as due to translational infidelity or lack of "other stuff." Sarcopenia or frailty appear to be highly correlated with vulnerability to several "age related" conditions [268].

Further, the impact of drugs common in the elderly on nutrient status is becoming recognized [19].

Atrophic gastritris is one form of digestive problem. It may be silent, associated with non-acid reflux, and lead directly to deficiencies of B-12 and iron [214].

The list of effected nutrients may be much larger however due to simple lack of data. Vitamin C destruction, rather than simple malabsorption, may be significant with increased stomach pH as well as calcium salt dissolution and absorption rates [39]. Presumably other cations than iron and calcium are impacted with copper being an important candidate likely to be ignored. Some studies suggest elevated serum levels with H pylori positive gastritis [10] which may be expected as Ceruloplasmin is an acute phase protein and blood levels may reflect an immediate stress. Generally malabsorption of copper or excess zinc was described in 2008 as uncommon but increasingly recognized [37].

The kinds of problems then include

Thinking outloud
catagory issue- ?

III.1. GI chemistry

III.2. incomplete digestion

III.3. neutralization of any possible ingested toxins.

III.4. insoluble formation, solubility enhanacement

III.5. microbial or spontaneous toxin formation

III.6. GI surface quality and transporter expression

Transporter expression needs to look at age related splice variants that AI suggests are a huge problem. Intriguingly, Hutchinson Gilfor progeria is thought to be caused by a genetic defect in splicing [144].

IV. MEAL ENGINEERING CONSIDERATIONS

Based on the above, the goal then is to replace nutrients and remove toxics that are more common in old age. Return to a youthful GI tract may be a reasonable goal although it may be possible to do even better. Conceivably if nutrient uptake can overcome GI limitations some healing may be observed making possible a return to normal food for a while.

Nutrients need only be supplied over "relevant" time scales. This can vary a lot from nutrient to nutrient and allows better segregation and rotation than trying to get everything everyday. Incompatibilities include chemistry and competition for receptors or enzymes or transporters etc.

Not every meals needs to be the same and in fact currently with the dogs one meal contains most things while a second is largely reserved for copper.

IV.1. Reduced toxins : chloride and benzoate

IV.2. Solubility : chloride, protons, phosphate, emulsifiers

IV.3. Free nutrients

V. CANDIDATE INGREDIENTS

Candidates are drawn from prior thoughts on food and vitamin supplements along with the presumed GI disturbances common in old age.

Any foods or supplements reputed to improve some age related condition could be examined. In many cases the folklore appears to be a useful starting point as it may rationalize well with the above theory or be supported by in-house work here with dog feeding.

One recent work on pre-digested food for groups suc as the elderly gives list of some ideas [262].

Whey peptides improved some functions in healthy people thought to be due to Trp-Tyr peptides [131].

V.1. fermented foods

Fermented foods meet many of the criteria for obtaining nutrients with age impaired digestions. They tend to have low pH, although a group of alkaline fermentation products is known [193] , and can include pH as part of the definition, and essential molecules such as free amino acids and vitamin k. They seem to be associated with reduced dementia risk [202]

V.2. olive and oleic acid and essential fatty acids

Olive oil (and more recently canola oil) has been an important part of in-house dog diets (manuscript in preparation) as well as being a highlighted component of the Mediterranean Diet. A variety of possible causal pathways could be imagined for it but here it is considered somewhat uniquely, along with lecithin and phosphorous sources, as an absorption enhancer. Its interesting to note, although of unknown significance, that yeast with a higher amount of membrane oleic acid have higher ethanol tolerance [271]. As part of a meal formulation it may help with uptake of lipophilic nutrients. Canola oil is thought to provide additional essential fatty acids.

V.3. choline, DES, and chloride

While the utility of choline in the brain has been well considered its role in digestion may not be fully appreciated. Pathological metabolites such as TMAO are known

but a lot of work also exists on solvent technology relevant to absorption from the GI tract.

There is debate about how the physico-chemical properties of the GI tract function in health and disease but "natural deep eutectic solvents(NADES) have been proposed to be part of this overall system [251].

Thinking outloud
the above notes incorporated water a
changes in hygroscopic features actu
gas.

Various tests have been done with solvents containing choline or other GI compatible molecules, typically some class of deep eutectic solvent, for enhancement of drug uptake [183] [70] [187] with some successes. One study to determine if a choline chloride-malic acid NaDES can improve uptake of various classes of hydrophobic drugs and some success was shown with basic and nautral but not acidic samples [222]. i

Interestinly, DES components may include common nutrients or spices such as arginine and citric acid [76].

Its interesting to note that while choline chloride was being explored under non-physiological conditions (120 C) for delignification of biomass (something similar to digestion of natural foods) chloride was found to be the only important component with NaCl working just as well [233] despite all the interest in deep eutectic solvents etc. Chloride was also thought to be reasonable for copper extraction compared to other biocompatible chemistries [165].

Thinking outloud
digestion section ?

Other techniques to improve absorption of hydrophobics that work in the GI tract include a SMEDDS system with added pi stacking [3]

2 related objectives are to improve chloride levels and reduce pH. In the results cited previously, acid came from citric acid and HCl supplement formulations. Formulations that use HCl as a raw component would require great care and may be avoided initially. Some soft drinks such as diet cola beverages contain phosphoric acid that may have some benefits. Chloride was largely from potassium chloride and again added HCl.

Title

Ingredient	Amount	InCompatible	Remarks
Essential Amino Acids			
Essential Fats			
Essential Metals			
B-vitamins			
Misc vitamins			
Ion control			
Digestion Improvement			
Microbiology/Toxin Control			

TABLE I. Catagories are never clean since small molecules always do many things and duplicates will be included

VI. DEVELOPMENT PATHS

Since many candidate components are GRAS, its possible to outline informal or "crowd sourced" data from motivated parties such as caregivers. Concerns about malicious data due to various incentives as well as other issues would have to be considered in detail. I have been using MUQED daily to record dog diets and relate to outcomes. This helped to identify olive oil as an improtant component although it may have a long response time if other sources of oleic acid such as chicken are included in the diet. The added workflow just requires a few minutes after each meal or dosing to enter whatever was given and have it checked for basic syntax. I use "vi" although more sophisticated or voice entry may be easy to do. The current implementation of MUQED creates a small vocabulary of foods, supplements and outcomes as well as a simply syntax for entering amounts right after or even skeletons before consumption. As the current implementation is designed for just "a few" people or dogs, everything is done with text files locally although collation with a standard vocabulary could be done with better scaling implementations.

My in-house results with dogs does suggest some folk lore or anecdotes may have useful signals but will require more data to sort out and a MUQED-like approach may fill the void.

Even if this fails to mitigate disease, it may not be a frivolous effort especially if any trends are observed with prescription medicines.

VII. BIGGEST PROBLEMS

Thinking outloud
lot of issues here include off-target effects, accurate prescribing patterns, population differences between users and non uers, any real effect etc. AGA seems to be pushing this things probably get ad revenue lol

As proton concentration is thought to be a big issue, it may be surprising that chronic PPI use does not lead to obvious epidemic of AD. Although in older people they may not have much real impact as stomach acid is already likely to be low (if PPI's are effective at anything it is likely through some other mechanism. If you can't tolerate stomach acid you are going to be malnourished anyway - something is wrong so you may expect accurately prescribed PPI users to already be predisposed to dementia). For H pylori control, it turns out they likely prevent a bacterial pump rather than a host pump from functioning leading to reduction [24]. In general they lead to less healthy microbioome and over abundance of Streptococcus that may also produce symptoms in patients [179]. Its worth noting that some species can secrete virulence factors inducible by a return to low pH [67]. The PPI's may appear to relieve symptoms but allow the organisms to proliferate making the situation worse with a return to physiological acid levels. (lol).

Current evidence is conflicting even among older or sick people who may already have GI impairments minimizing any effect of PPI's. A 2023 study of a 1.98 million Danish cohort found an incidence of any dementia ratio between PPI users and non-users of 1.36 but reducing with age [203]. The American Gastroenterological Association appears to advance the position that this is unlikely. One post hoc analysis of the ASPREE aspirin trial involving people older than 65, found no association between PPI use and dementia [176]. Reported results are heavily corrected for various factors and the raw HR was not immediately obvious. It is possible the low dose aspirin has some impact but difficult to tell as presented.

Protons and chloride appear to be differentially regulated and PPI's do not stop normal chloride secretion [54] despite the fact they may be used to prevent excess fecal chloride excretion in abnormal cases [5]. There are several chloride transporters in the stomach including calcium activated [211] that may decouple from PPI's. The cystic fibrosis literature may provide some clues.

However, the theory here is a bit empirical and recognizes other factors than pH alone may be important. With a non-zero supply of most required molecules adaptive responses may be able to move the effects of the deficiencies around making the clinical presentation difficult to trace to original problem.

Even if the theory is sound and eventually is proven useful as a starting point, social issues are a big problem. For whatever reason amyloid drugs appear to be getting a more favorable response than is supported by known data. Anything involving skin pigment will attract angry people.

VIII. CONCLUSIONS

The present work seeks to find a robust intvention against AD, specifically LOAD but more generally most age related conditions, by engineering meals to mitigate or fix GI impairments that can reduce nutrient uptake and increase toxin production. It suggests that the sum of all the partial theories can best be described by this common source of failure or it identified the proverbial elephant from distinct observations of the pieces. One non-specific symptom is likely weak structural proteins as amino acids and various factors will be deficient. While several paths for investigation are possible, informal approaches by motivated parties may help avoid some problems of the past efforts such as clinging to a familiar target instead of following the evolving data and avoiding ill posed questions such as the action of a single molecular entity on a diverse randomized population instead of the components for even a single cure specialized to even one person. As a corollary, it may be that certain identifiable groups are more susceptible to mental or physical injuries due to inappropriate meal design and attempts to just blame undefined social pressures will be counterproductive for everyone.

IX. SUPPLEMENTAL INFORMATION

Example of MUQED output on dogs over several years, `https://github.com/mmarchywka/dogdata` and some MUQED source code `https://github.com/mmarchywka/muqed_util`

IX.1. Computer Code

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ACKNOWLEDGMENTS

1. Pubmed eutils facilities and the basic research it provides.
2. Free software including Linux, R, LaTex etc.
3. Thanks everyone who contributed incidental support.

Appendix A: Statement of Conflicts

No specific funding was used in this effort and there are no relationships with others that could create a conflict of interest. I would like to develop these ideas further and have obvious bias towards making them appear successful. Barbara Cade, the dog owner, has worked in the pet food industry but this does not likely create a conflict. We have no interest in the makers of any of the products named in this work.

Appendix B: About the Authors and Facility

This work was performed at a dog rescue run by Barbara Cade and housed in rural Georgia. The author of this report ,Mike Marchywka, has a background in electrical engineering and has done extensive research using free online literature sources. I hope to find additional people interested in critically examining the results and verify that they can be reproduced effectively to treat other dogs.

Appendix C: Symbols, Abbreviations and Colloquialisms

TERM	definition and meaning

Appendix D: General caveats and disclaimer

This document was created in the hope it will be interesting to someone including me by providing information about some topic that may include personal experience or a literature review or description of a speculative theory or idea. There is no assurance that the content of this work will be useful for any paricular purpose.

All statements in this document were true to the best of my knowledge at the time they were made and every attempt is made to assure they are not misleading or confusing. However, information provided by others and observations that can be manipulated by unknown causes (”gaslighting”) may be misleading. Any use of this information should be preceded by validation including replication where feasible. Errors may enter into the final work at every step from conception and research to final editing.

Documents labelled ”NOTES” or ”not public” contain substantial informal or speculative content that may be terse and poorly edited or even sarcastic or profane. Documents labelled as ”public” have generally been edited to be more coherent but probably have not been reviewed or proof read.

Generally non-public documents are labelled as such to avoid confusion and embarassment and should be read with that understanding.

Appendix E: Citing this as a tech report or white paper

Note: This is mostly manually entered and not assured to be error free.
This is tech report MJM-2026-001.

Version	Date	Comments
0.01	2026-01-17	Create from empty.tex template
-	March 30, 2026	version 0.00 MJM-2026-001
1.0	20xx-xx-xx	First revision for distribution

Released versions,
build script needs to include empty releases.tex

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Supporting files. Note that some dates,sizes, and md5's will change as this is rebuilt.

This really needs to include the data analysis code but right now it is auto generated picking up things from prior build in many cases

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